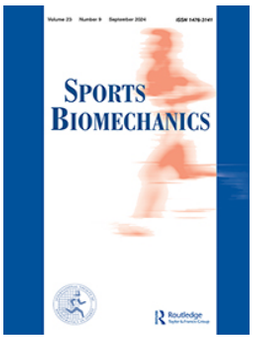

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Dynamic adjustment of submaximal effort soccer side-foot kicks

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ABSTRACT

We aimed to illustrate kicking leg dynamics during submaximal effort soccer side-foot kicks. Side-foot kicks with three effort levels (50, 75 and 100% effort levels based on maximal effort) of eight male university soccer players were captured (500 Hz) while initial ball velocities were monitored simultaneously. Systematic regulation in joint kinetics (angular impulses) was clearly demonstrated for hip flexion and knee extension moments thereby supporting the interpretation that the final foot velocity is controlled in a context of a planar, sequential segmental system. Out of the thigh-shank plane motion (hip external rotation moment) was also found to be systematically adjusted. Kinematic contributions of knee extension angular velocity to the final foot velocity increased significantly in the maximal effort while that of hip external rotation reduced significantly, coinciding with a more straightforward approach-run. The adjustable range of the foot-ball interaction was found to be rather smaller in side-foot kicks. However, significantly smaller ball/foot velocity ratios in the two submaximal conditions suggested ankle joint fixation was manipulated towards ball impact. Players and coaches ought to recognise that the intensities of side-foot kicks were regulated by the motions within and without the thigh-shank plane alongside several kinematic changes.

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Side-foot kick; association football; submaximal effort; inverse dynamics

Introduction

In the course of football (soccer) matches, players are demanded to use various types of kicking techniques. Among numerous types of kicks used in soccer, the side-foot kick is the most frequently used technique when a shorter and more precise pass or shot is required (Nunome et al., 2002). This claim was confirmed by Mitschke and Milani (2014). They reported that soccer players used the side-foot kick to complete the majority number of passes (59.6%) and mainly used it for low/short passes in high-level soccer competitions.

While much is known about the biomechanics of the instep kick (Asami & Nolte, 1983; Barfield, 1995; Dörge et al., 2002; Isokawa & Lees, 1988; Lees & Nolan, 1998; Lees at al., 2010; Nunome et al., 2006a, 2006b; Roberts & Metcalfe, 1968; Rodano & Tavana,

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1993), a relatively smaller number of studies (Brophy et al., 2007; Kawamoto et al., 2007; Levanon & Dapena, 1998; Nunome et al., 2002; Zago et al., 2014) focused on the biomechanics of the side-foot kick, to date. In these previous attempts, except for one study (Zago et al., 2014), all the studies investigated maximal effort trials in terms of kinematics (Brophy et al., 2007; Levanon & Dapena, 1998), muscle recruitment patterns (Brophy et al., 2007) and kinetics (Kawamoto et al., 2007; Nunome et al., 2002).

During a soccer match, players are often required to kick the ball with a wide range of velocities (Hanson et al., 2012). Although there have been a few attempts to illustrate kinematic aspects (Andersen & Dörge, 2011; Scurr & Hall, 2009; Teixeira, 1999) or muscle recruitment patterns of selected lower limb muscles (Katis et al., 2013; Scurr et al., 2011; Zago et al., 2014) of soccer kicks in accuracy enhanced submaximal effort conditions, the information of the dynamics of the submaximal kicking motion has been limited, except the study of Nunome et al. (2018). To the best of our knowledge, the study of Nunome et al. (2018) was the first that systematically controlled the effort levels of soccer instep kicks to provide novel findings regarding leg swing regulations from the viewpoint of dynamics. However, this feature of the side-foot kick has been under-researched, to date.

As the side-foot kick is the most frequently used technique for passing (Mitschke & Milani, 2014) during matches, it is reasonable to assume soccer players are required to execute the side-foot kicking to provide passes with varying intensities. Thus, similar to the instep kick, it can be expected that examining the dynamics of submaximal effort side-foot kicking would provide in-depth insight as to how players adjust the intensity of this kicking technique in a match situation. Identifying key motions that are responsible for adjustments in side-foot kicking intensity would be useful, practically relevant, and instructive information for coaches. Based on the information, coaches may instruct their players on how to control the intensity of the side-foot kicking technique depending on situations likely to happen in an actual course of a match.

To date, two previous studies reported that several out of the thigh-shank plane motions form the foundation of the side-foot kick motion. Levanon and Dapena (1998) through their kinematic analysis, highlighted the importance of hip adduction motion which adds the velocity component normal to the thigh-shank plane allowing players to hit the ball squarely with the medial side of the foot. In addition, there were small but positive contributions of hip adduction (8%) and external rotation (9%) to the forward velocity foot centre of gravity (CG) immediately before ball impact. In contrast, Nunome et al. (2002) demonstrated another style of side-foot kicks, in which players utilise hip external rotation moment to increase directly the forward velocity of the medial side of the foot. Thus, players are expected to systematically control these out-of-plane motions to meet submaximal demands for the side-foot kicks.

We aimed to illustrate the dynamics of the kicking leg during soccer side-foot kicks at several submaximal effort levels to gain our understanding of how players change (adjust) the dynamics in response to the demands of the task (decreased ball velocity). Nunome et al. (2018) found for the instep kick that in response to three target effort levels, precise adjustments occurred in angular impulses of the hip flexion and the knee extension moments in a context of proximal to distal segmental sequence during soccer instep kicking. It can be expected that a similar pattern of adjustment should be achieved in the side-foot kick. Moreover, as Levanon and Dapena (1998) and Nunome et al. (2002)

demonstrated that hip adduction and external rotation motions play a substantial role to swing the kicking foot forward during side-foot kicking, it can be reasonable to expect that moments regarding these two motions would also be regulated systematically to meet submaximal demands. With regard to these previous findings, we set two hypotheses: 1) similar to the instep kick, systematic adjustments will be conducted in the dynamics of sequential planar joint motions, and 2) players also regulate hip adduction and external rotation moments to achieve the necessary submaximal ball velocity.

Methods

Eight male university-level soccer players (age = 20.4 ± 1.5 years; height = 172.4 ± 4.6 cm; body mass = 63.3 ± 5.2 kg; soccer experience = 13.1 ± 3.1 years) who regularly compete in the second division of a regional university league in Japan volunteered to participate in the present study. The present study was conducted in accordance with the guidelines of the Declaration of Helsinki. Informed written consent was obtained from each participant before the experiment. The experiment protocol was approved by the Human Research Committee of a university.

All participants were free from any lower limb injuries when they participated in the experiment and preferred to kick the ball with their right leg. After a warm-up session which was similar to their usual preparation for training sessions (~15 mins including jogging, stretching, and kicking of increasing effort), participants performed soccer side-foot kicks with three effort levels, towards the centre of a target (88 × 88 cm) set in the middle of a small indoor soccer goal 7 m ahead. The bull's eye of the target was set at a height of 1 m from the ground. The three effort levels were defined by the resultant initial ball velocity on an individual basis. After a few times practice trials, the participants were asked to perform their maximal effort kick twice before the trials to obtain the baseline of target ball velocity (100% effort). The maximal ball velocity of each participant was predetermined by the kicked ball motion, which was captured using a 10-camera optoelectronic motion capture system (Vicon Nexus; Vicon Motion Systems, Oxford, UK). From the average ball velocity of these two maximal trials, target ball velocities of 75 and 50% effort levels were determined for each individual. The participants were then asked to perform the side-foot kicks in three effort levels (50, 75 and 100%) in a randomised order. The participants approached the ball using two steps from their preferred approach angles. To avoid the possible influence of fatigue, the participants recovered for at least 1 min between trials. During the kicking trials, kicked ball velocities were monitored by a pair of photocells, which aligned towards the goal with a 1 m interval and fed back to the participants immediately after each trial. Based on the information, the participants are allowed to adjust their effort levels to achieve their target velocities. All the participants repeated the kicks at the three effort levels to obtain three 'good' shots that satisfied the criteria (±5% of the target velocity and hitting adequate centre targeting in the goal (hit the target)).

The motions of the kicking leg, pelvis, and ball were three-dimensionally captured simultaneously at 500 Hz using the motion capture system. The participants wore the same type of soccer shoes for indoor use although differing in size. The detailed procedures including marker set on the body and ball are described in the study of Inoue et al. (2014). Resultant joint moments and joint angular velocities were computed

using the procedure of Nunome et al. (2002). Angular impulses were computed as the summation of resultant joint moments. Angular impulse components: hip flexion, knee extension, hip adduction, and hip external rotation were summed using the trapezoidal rule throughout the kicking phase defined from the moment of the kicking leg toe-off to ball impact (Nunome et al., 2002). Segment mass, the centre of the mass location and moment of inertial values were derived from the data of young Japanese athletes (Ae et al., 1992), which was the most appropriate for the participants in the present study. The kicking leg was modelled as a two-link kinetic chain composed of the thigh and lower leg (Nunome et al., 2006a), assuming the foot joint motion has no substantial influence on the leg swing motion. Prior to modelling the lower leg segment, the mass of the shoe was added to the foot mass, and its effect on the length of radius gyration of the foot segment was assumed to be small enough to be negligible. To exclude artefacts caused by smoothing through ball impact, the smoothing procedure defined by Nunome, Ikegami, et al. (2006a) was applied. Following the procedure, all kinematic and kinetic variables of the kicking leg were first computed from raw coordinates and then extrapolated and filtered using a fourth-order Butterworth filter at 12.5 Hz.

Initial ball velocities, final foot CG velocities immediately before ball impact, ball/foot velocity ratios, angular impulses due to the moments and joint angular velocities were calculated for each trial (a total of 72 kicks). Three trials in each condition were averaged for each participant and then the values were used to compute the discrete mean, SD and ensemble average curves. The initial ball velocities in the consecutive trials were recalculated from the trajectory of the geometrical centre of the ball, estimated from eight ball surface makers using the least square method (Inoue et al., 2014). One-way ANOVA with repeated measures was used for comparisons between effort levels. The statistical significance was set at $P < 0.05$. Multiple comparisons using the Bonferroni adjustment were used for post hoc testing (t-test) whenever the ANOVA returned a significant result. Adjusted probability values less than 0.05 were judged to be significant. Effect sizes were also estimated as Cohen's D and evaluated as trivial (0–0.19), small (0.20–0.49), medium (0.50–0.79), and large (0.80 and greater).

Results

Significant differences with large effect size ($P < 0.001$) were evident for initial ball velocity and final foot velocity between each effort level while the difference in ball/foot velocity ratios between 75% and 50% of effort levels was insignificant. Also, ball/foot velocity ratio at 100% effort level was significantly higher with a large effect size than those in the two submaximal effort levels (Table 1). On average, the participants in the present study used 77.6% and 58.0% of the relative final foot CG velocity (per 100% effort) for producing 75.8% and 53.3% of the relative initial ball velocities (per 100% effort), respectively.

Apparent kinetic and kinematic differences of soccer side-foot kicks were demonstrated between each effort level for planar motions within the thigh-shank plane: the hip flexion moment and the knee extension moment (Figure 1). Angular impulses due to these moments were found to be systematically reduced in lower effort levels and significantly different ($P < 0.001$) with a large effect size between each effort level (Table 2). Peak hip flexion angular velocities were systematically changed and increased

Table 1. Average (±SD) values of the velocity of ball and foot centre of gravity (CG) and foot CG—ball velocity ratio.

	Effort level			ANOVA	Post hoc100% vs 75%	Post hoc100% vs 50%	Post hoc75% vs 50%
	100%	75%	50%	*p* value (partial η^2)	*p* value (Cohen's d)	*p* value (Cohen's d)	*p* value (Cohen's d)
Initial ball velocity (m/s)	23.6 (1.4)	17.9 (1.3)	12.6 (1.0)	<0 (0.990)*	<0 (4.0)*	<0 (8.5)*	<0.001 (4.3)*
Final foot velocity (m/s)	17.4 (0.8)	13.5 (0.9)	10.1 (1.2)	<0 (0.984)*	<0 (4.4)*	<0 (6.8)*	<0.001 (3.1)*
Foot-ball velocity ratio	1.35 (0.04)	1.32 (0.03)	1.25 (0.07)	0.015 (0.593)*	0.046 (0.8)*	0.038 (1.7)*	ns (1.2)

*: Significant difference in post hoc comparison with Bonferroni adjustment ($p < 0.05$).

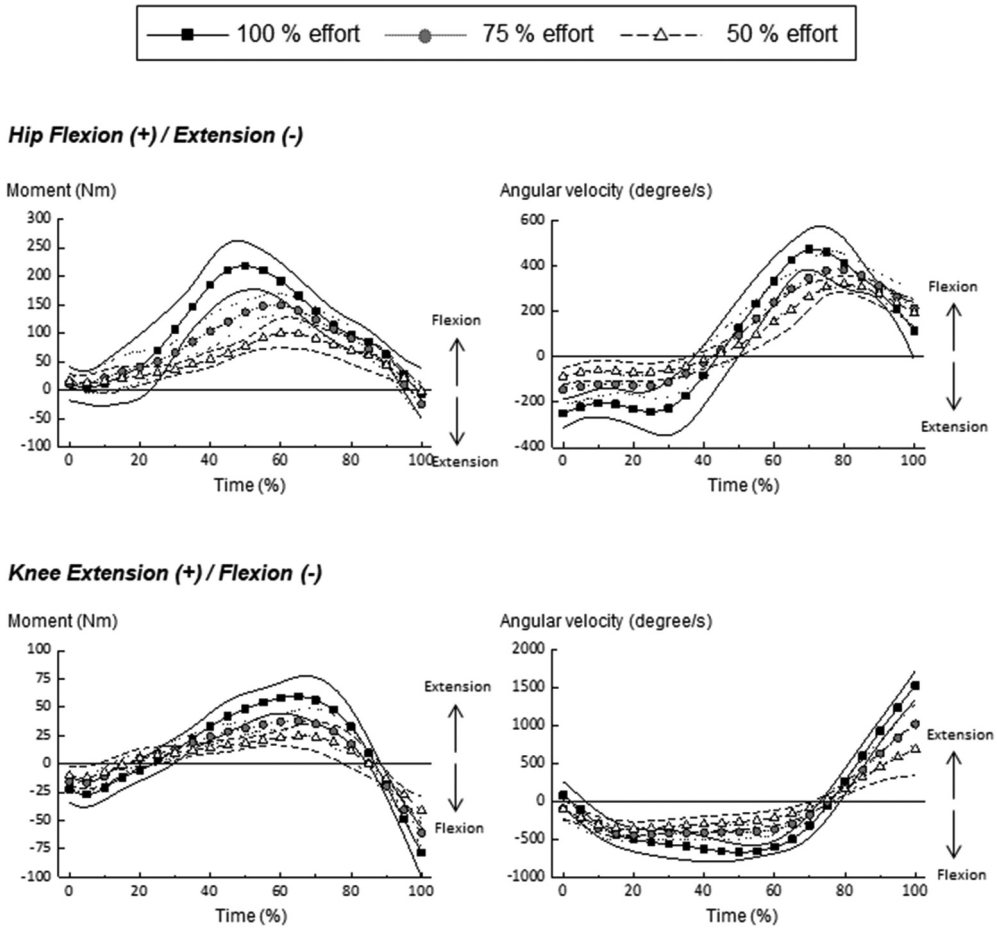

Figure 1. Average (±SD) changes of joint moments and joint angular velocities for motions within thigh-shank plane: hip flexion/extension (top panels) and knee extension/flexion (bottom panels) during soccer side-foot kicks in three effort levels (50%, 75% & 100%).

significantly (100% vs. 75%, $P < 0.001$; 75% vs. 50%, $P < 0.05$) with a large effect size. Peak knee extension angular velocities also showed a systematic, significant increase ($P < 0.001$) as the effort level increased with a large effect size.

Regarding out of the thigh-shank plane motion, it can be seen that the hip external rotation moment and the hip adduction moment were consistently exhibited in all the effort levels and their angular velocities were apparent during kicking (Figure 2). Angular impulses due to the hip external rotation moment were systematically changed and significantly ($P < 0.001$) increased with a large effect size between each effort level. In contrast, although angular impulses of the hip adduction moment at 75% effort level increased significantly ($P < 0.01$) than that of 50% effort level with a moderate effect size, those in two submaximal effort levels (75% and 50%) did not differ significantly compared to that of maximal effort level (100%). It is interesting to note, while angular impulses due to the hip external rotation moment were systematically regulated, angular velocities of the hip external rotation did not match with this systematic change (Table 2).

Table 2. Average (±SD) values of angular impulses due to resultant joint moments and joint angular velocities for selected joint motions.

	Effort level			ANOVA	Post hoc 100% vs 75%	Post hoc 100% vs 50%	Post hoc 75% vs 50%
	100%	75%	50%	p value (partial η^2)	p value (Cohen's d)	p value (Cohen's d)	p value (Cohen's d)
Thigh–shank plane motion							
Angular impulse (Nm.s)							
Hip flexion	25.5 (2.7)	19.5 (2.5)	14.7 (3.5)	<0 (0.956)*	<0 (2.2)*	<0 (3.3)*	<0 (1.5)*
Knee extension	5.8 (0.9)	4.1 (0.7)	3.0 (1.0)	<0 (0.898)*	<0 (1.9)*	<0 (2.7)*	<0 (1.2)*
Peak angular velocity (degree/s)							
Hip flexion	500.1 (89.4)	407.9 (76.9)	335.4 (41.7)	<0 (0.794)*	0.004 (1.0)*	0.001 (2.2)*	0.03 (1.1)*
Knee extension	1529.3 (192.1)	1022.7 (258.4)	697.3 (344.8)	<0 (0.916)*	<0 (2.1)*	<0 (2.8)*	<0 (1.0)*
Out of plane motion							
Angular impulse (Nm.s)							
Hip adduction	7.4 (3.8)	6.5 (2.7)	4.7 (1.9)	0.03 (0.533)*	ns (0.3)	ns (0.8)	0.010 (0.7)*
Hip External rotation	5.9 (0.9)	4.2 (0.9)	3.0 (1.0)	<0 (0.948)*	<0 (1.7)*	<0 (2.9)*	<0 (1.3)*
Peak angular velocity (degree/s)							
Hip adduction	263.4 (97.0)	180.1 (78.8)	130.6 (45.4)	0 (0.594)*	ns (0.9)	0.02 (1.7)*	ns (0.7)
Hip External rotation	520.9 (176.2)	658.8 (255.0)	516.1 (213.9)	0 (0.555)*	ns (0.6)	ns (0.0)	0.01 (0.6)*

*: Significant difference in post hoc comparison with Bonferroni adjustment ($p < 0.05$).

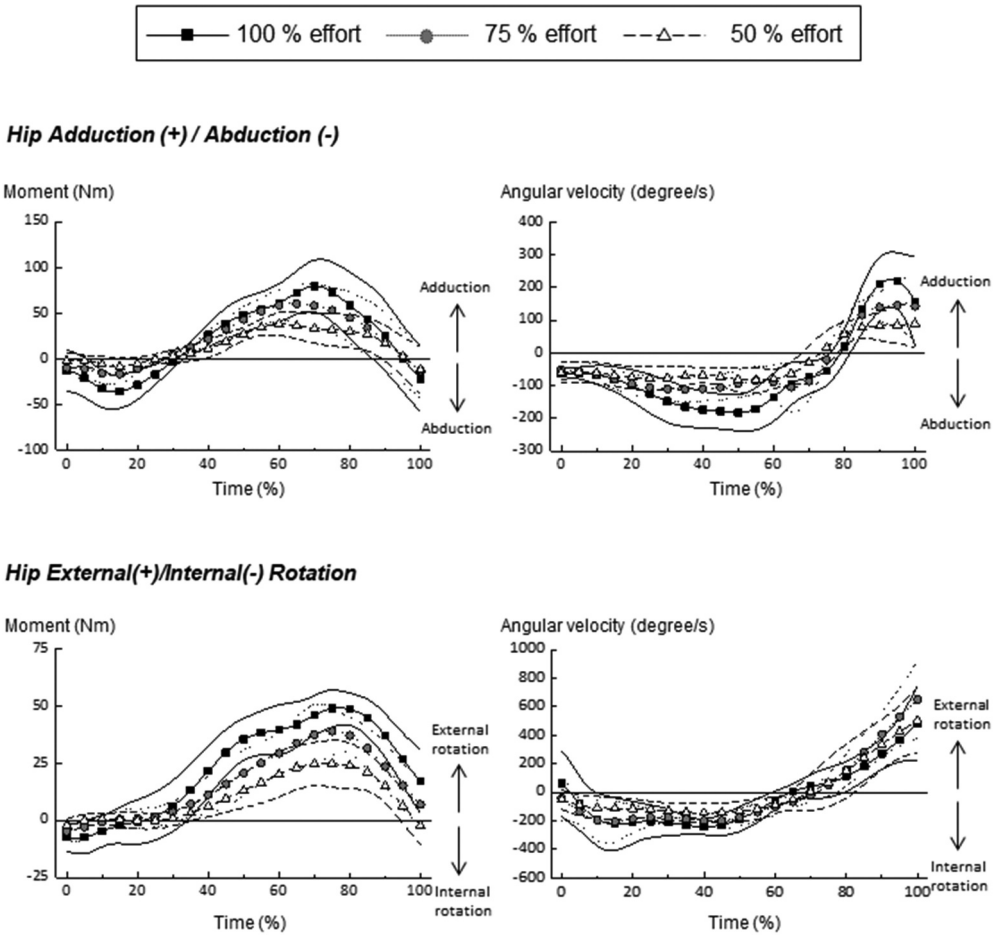

Figure 2. Average (±SD) changes of joint moments and joint angular velocities for out of thigh-shank plane motions: hip adduction/abduction (top panels) and hip external/internal rotation (bottom panels) during soccer side-foot kicks in three effort levels (50%, 75% & 100%).

On average, trials of 75% effort level showed the highest average peak angular velocity among the three effort levels and did not significantly differ from that of the maximum effort level (100%). For the angular velocities of the hip adduction, a significant difference was solely observed between 50% and 100% effort levels with a large effect size (Table 2).

Discussion and implication

In the present study, we aimed to clarify the differences in soccer side-foot kick dynamics between three effort levels (50%, 75%, and 100%). For planar motions within the thigh-shank plane, kinetic adjustments were apparent, in which angular impulses due to both the hip flexion and the knee extension moments were precisely regulated to meet the submaximal demands of kicking. This finding was consistent with that in soccer instep kick (Nunome et al., 2018), thereby fully supporting our first hypothesis. On the other hand, as the side-foot kick includes out-of-the-plane motions to swing the medial side of

the foot forward (Levanon & Dapena, 1998; Nunome et al., 2002), we expected that those motions are also regulated in the same manner. Although angular impulses due to the hip external moment were systematically regulated, those due to the hip adduction moment did not follow a systematic manner of regulation. Thus, our second hypothesis was not fully supported by the finding regarding hip adduction motion.

Similar to the instep kick (Nunome et al., 2018), dynamics adjustments were clearly identified in both proximal and distal joints: the hip flexion and knee extension within the thigh-shank plane. As shown in Figure 1, there are apparent regulations of the hip flexion and knee extension moments during kicking. Angular impulses due to these moments (areas under the curve) were rather precisely adjusted to target effort levels (Table 2). On average, the relative amount (per 100% effort) of angular impulses due to the hip flexion moment in two submaximal effort levels (75% & 50%) were 76.7% and 57.9% and those due to the knee extension moment were 71.5% and 52.5%, respectively. Gaps between the target effort levels and relative angular impulses become considerably smaller when these relative angular impulses are compared with the relative final foot CG velocities. In this comparison, the gaps between the relative hip flexion angular impulses and the final foot CG velocities fall within the sub % level, in which the present participants produced 77.6% and 58.0% of the relative final foot velocities using 76.7% and 57.9% of the relative hip flexion angular impulses, respectively. This finding in particular for the hip flexion moment strongly suggests that to control a series of planar motions, the moment plays a crucial role in regulating the effort levels of the side-foot kick. Thus, it is reasonable to interpret that intensities of soccer side-foot kicks are controlled in a context of a proximal to distal sequential system as well as for soccer instep kicks (Nunome et al., 2018).

Nunome et al. (2006a) demonstrated segmental dynamics during the instep kick, in which the hip flexion moment is most likely the exclusive kinetic source responsible for thigh segment counter clock-wise rotation (approximate hip flexion motion) while two kinetic sources (joint moment and interaction moment) dominate that of the lower leg segment (approximate knee extension motion). In addition, as Putnam (1991) reported, the interaction moment acting on the lower leg segment is dependent on the thigh angular motion (approximate hip angular motion). Thus, from these aspects in lower limb segmental dynamics, it is reasonable to consider that the hip flexion moment is most likely responsible for regulating the sequential planar motions within the thigh-shank plane. In contrast, it can be suggested that knee extension moment does not fully control knee extension motion. This may partially explain a more precise control exists in angular impulses due to the hip flexion moment. Players should be focused more on controlling the hip flexion motion when they need to adjust the intensity of side-foot kicking. Also, coaches should be aware that controlling the hip flexion motion indirectly affects the control of the knee extension motion. Based on this information, they may instruct their players that the hip flexion motion ought to be precisely regulated to meet submaximal demands of side-foot kicking.

Angular impulses due to the hip external rotation moment were systematically changed and significantly ($P < 0.001$) different between each effort level with a large effect size while those due to hip adduction moment in two submaximal effort levels did not differ significantly compared to that of maximal effort level (Table 2). These findings suggest that among out of the thigh-shank plane motions, the present participants are

more reliant on the hip external rotation moment to regulate the final foot velocity during submaximal effort side-foot kicks. However, the gaps between relative final foot velocities and the relative amount of angular hip external rotation impulses are rather larger than those of hip flexion angular impulses. On average, the present participants produced 77.6% and 58.0% of the relative final foot velocities using 71.9% and 50.5% of the relative hip external rotation angular impulses. In addition, the hip external rotation angular velocities did not associate with this systematic change of the three effort levels. On average, trials of 75% effort level showed the highest peak angular velocity among the three effort levels and the value did not differ significantly from that of the 100% effort level (Table 2). This inconsistent change may imply some kinematic change of kicking occurred in particular due to the contribution of hip external rotation to the leg swing. To examine this issue, kinematic contributions of joint angular velocity to the final foot CG velocity immediately before ball impact were computed using the procedure of Sprigings et al. (1994). Also, the angle of approach was computed because the approach angle might affect the contributions of various planar and out-of-plane kinematics parameters. The approach angle was defined as the angle between the vector defined from the middle position of the left and right ASIS at 200 ms before the support leg touchdown to that of the moment of the support leg touchdown and the global Y-axis. Table 3 summarises the results of these post hoc analyses. As expected, the contribution of hip external rotation at the maximum effort (100%) level becomes significantly smaller than that of 75% effort. This change coincided with a significantly smaller approach angle at the maximum effort level (100%) compared to the 75% effort level. In contrast to this change, the contribution of knee extension at the maximum effort (100%) level becomes significantly larger than those of other effort levels (75% and 50%). These results suggest that in the side-foot kick, the change in kicking intensity coincided with kinematic alternations including the angle of approach. In the present case, when the participants demanded maximal effort, they tended to approach the ball more straightforwardly and more rely on knee extension motion to maximise the kicking leg swing. As the knee extension angular velocity is evidently the fastest angular velocity among all the leg motions during kicking, it should be a reasonable adaptation for players to maximise the final foot velocity. Also, a more straightforward approach angle might allow the participants to have a feasible posture to maximise the contribution of knee extension motion. Players should recognise that they may regulate the intensities of side-foot kicking by controlling the hip external rotation motion, altering lower limb postures and approach angles. These changes might work for players to facilitate providing passes/shots with adequate intensities.

There is one study focused on submaximal efforts of soccer instep kicks (Nunome et al., 2018), in which differences between the final foot velocity and resultant ball velocity were even emphasised when players kick the ball with submaximal effort levels. During submaximal effort instep kicks, the players actually used 79.6% and 61.5% of the relative speeds of foot CG (per 100% effort) to produce 75.3% and 53.2% of the relative resultant ball velocities (per 100% effort), respectively. There was a clear tendency that they hit an off-centre part of the ball using a more medial side of the foot and with a less upright foot posture. This finding can be interpreted as some fine-tuning of the resultant ball velocity (4.3% in 75% effort; 8.2% in 50% effort) by changing the manner of ball impact with altered foot posture (similar to the side-foot kick) and off-centre ball impact. In the present study, the gaps between final foot and ball velocities in submaximal conditions seem rather

Table 3. Average (±SD) values of kinematic contribution of joint rotation to the forward velocity of foot centre of gravity (CG) immediately before ball impact, the angle of approach and effective striking mass used for ball impact.

	Effort level			ANOVA	Post hoc 100% vs 75%	Post hoc 100% vs 50%	Post hoc 75% vs 50%
	100%	75%	50%	P value (partial η^2)	P value (Cohen's d)	P value (Cohen's d)	P value (Cohen's d)
Contribution (%)							
Hip external rotation	25.7 (9.9)	30.1 (8.5)	29.5 (12.6)	0.048 (0.352)*	0.041 (1.154)*	0.328 (0.648)	1.000 (0.116)
Knee extension	54.8 (10.5)	43.9 (16.5)	39.2 (22.5)	0.009 (0.608)**	0.015 (1.418)*	0.037 (1.183)*	0.317 (0.657)
Others	19.5 (7.8)	26.0 (11.0)	31.2 (15.2)	0.046 (0.438)*	0.324 (0.651)	0.127 (0.875)	0.056 (1.076)
Approach angle (degree)							
	15.1 (9.6)	17.9 (10.1)	18.0 (9.8)	0.026 (0.407)*	0.014 (0.262)*	ns (0.280)	ns (0.013)
Striking mass (kg)							
	1.92 (0.22)	1.78 (0.17)	1.84 (0.24)	0.031 (0.392)*	0.043 (0.661)*	ns (0.319)	ns (0.273)

*: Significant difference in post hoc comparison with Bonferroni adjustment ($p < 0.05$).

smaller (1.8% in 75% effort; 4.7% in 50% effort). It can be expected that soccer players might have smaller margins to regulate the foot—ball interaction when they hit the ball with the medial side of the foot. To clarify this research question, the foot contact point on the ball and foot orientation at the moment of ball impact was computed as a post hoc analysis and compared between each effort level as is done for the instep kick (Nunome et al., 2018). Figure 3 highlighted the differences in the ball impact manner of the side-foot kick to meet lower demands. As shown, the foot contact points on the ball (top panel) and foot postures (bottom panels) overlapped in most parts among the three effort levels. Evidently, except one participant showed a unique foot posture at the maximal effort (100%), the rest of the present participants tended to repeat a kinematically similar manner of ball impact in either effort level, suggesting they could have a different adjusting strategy

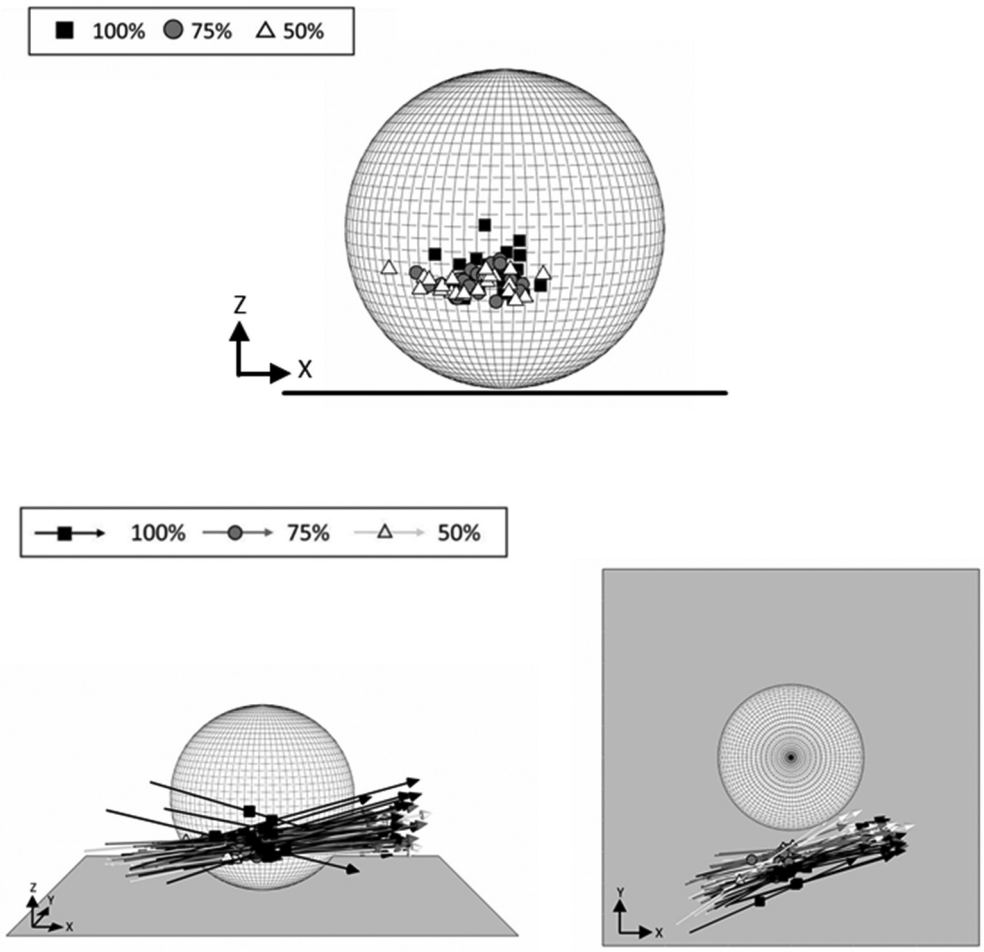

Figure 3. The back view of foot centre of gravity contact point on the ball at the instant of ball impact (top panel) and foot orientations at the instance of ball impact (bottom panels). The left bottom panel shows the back view and the right bottom panel represents the top views of the foot orientation at the instant of ball impact.

than that of the instep kicks. As expected, there might be a smaller adjustable range of the foot-ball interaction to meet lower demands when players use the side-foot kicks.

In contrast to the relatively smaller adjustable range of the foot-ball interaction during side-foot kicking, ball/foot velocity ratios in the two submaximal levels (75% & 50%) reduced significantly compared to that of the maximal (100%) effort level (Table 1). This finding suggests that players might add some fine-tunings to the resultant ball velocity by changing the manner of ball impact other than the appreciable way observed in the instep kick (Nunome et al., 2018). One possible strategy is to adjust the foot segment fixation to the proximal segment. Peacock and Ball (2018), although the authors adapted a kicking style similar to the instep kick, used a kicking machine to clarify the influence of joint rigidity on impact efficiency and succeeded in demonstrating that ankle joint fixation significantly increased the effective striking mass with a large effect. To estimate the mass used for ball impact, the effective striking mass (Asami & Nolte, 1983) was computed in each effort level (Table 3). In either effort level, the effective striking mass apparently exceeded the estimated foot segment mass with the shoe (1.30 ± 0.58 kg), and the striking mass at 75% effort level was found to be significantly smaller than that of 100% effort level with a medium effect. This finding suggests that the level of ankle joint fixation may be responsible for the difference in the foot-ball interaction between these two effort levels.

Conclusion

In the present study, we aimed to illustrate how soccer players regulate the side-foot kick intensities to meet submaximal demands. Similar to the instep kicks, submaximal side-foot kicks were characterised by significantly different angular impulses due to the hip flexion and the knee extension moments between each effort level. Of these, the hip flexion moment was found as the key moment to regulate the intensity of the side-foot kick because its angular impulse showed the most precise regulation to meet submaximal demands. Regarding out of the thigh-shank plane motions, angular impulses due to the hip external rotation moment were found to be significantly different between each effort level and adjusted in proportion to the three effort levels, suggesting this moment is another key systematically regulating the side-foot kick intensities. Finally, in contrast to the instep kick, players tend to hit the ball in a similar kinematic manner regardless of the effort levels. However, as ball/foot velocity ratios varied significantly between the maximal and the two submaximal conditions, possible manipulation of ankle joint fixation to alter ball impact efficiencies was suggested.

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References

Ae, M., Tang, H., & Yokoi, T. (1992). Estimation of inertial properties on the body segments in Japanese athletes. In Society of Biomechanisms Japan (Eds.), *Biomechanisms 11: Form, motion, and function in humans* (pp. 23–33). The University of Tokyo Press.

Andersen, T. B., & Dörge, H. C. (2011). The influence of speed of approach and accuracy constrain on the maximal speed of the ball in soccer kicking. *Scandinavian Journal of Medicine & Science in Sports*, *21*(1), 79–84. https://doi.org/10.1111/j.1600-0838.2009.01024.x

Asami, T., & Nolte, V. (1983). Analysis of powerful ball kicking. In H. Matsui & K. Kobayashi (Eds.), *Biomechanics VIII-B* (pp. 695–700). Human Kinetics.

Barfield, W. R. (1995). Effects of selected kinematic and kinetic variables on instep kicking with dominant and nondominant limbs. *Journal of Human Movement Studies*, *29*, 251–272.

Brophy, R. H., Backus, S. I., Pansy, B. S., Lyman, S., & Williams, R. J. (2007). Lower extremity muscle activation and alignment during the soccer instep and side-foot kicks. *Journal of Orthopaedic & Sports Physical Therapy*, *37*(5), 260–268. https://doi.org/10.2519/jospt.2007.2255

Dörge, H. C., Andersen, T. B., Sørensen, H., & Simonsen, E. B. (2002). Biomechanical differences in soccer kicking with the preferred and the non-preferred leg. *Journal of Sports Sciences*, *20*(4), 293–299. https://doi.org/10.1080/026404102753576062

Hanson, H., Harland, A., Holmes, C., & Lucas, T. (2012). Method for understanding football ball motions using video-based notational analysis. *Procedia Engineering*, *34*, 164–169. https://doi.org/10.1016/j.proeng.2012.04.029

Inoue, K., Nunome, H., Sterzing, T., Shinkai, H., & Ikegami, Y. (2014). Dynamics of the support leg in soccer instep kicking. *Journal of Sports Sciences*, *32*(11), 1023–1032. https://doi.org/10.1080/02640414.2014.886126

Isokawa, M., & Lees, A. (1988). A biomechanical analysis of the instep kick motion in soccer. In T. Reilly, A. Lees, K. Davies, & W. J. Murphy (Eds.), *Science & Football* (pp. 449–455). E & FN Spon.

Katis, A., Giannadakis, E., Kannas, T., Amiridis, I., Kellis, E., & Lees, A. (2013). Mechanisms that influence accuracy of the soccer kick. *Journal of Electromyography and Kinesiology*, *23*(1), 125–131. https://doi.org/10.1016/j.jelekin.2012.08.020

Kawamoto, R., Miyagi, O., Ohashi, J., & Fukashiro, S. (2007). Kinetic comparison of a side-foot soccer kick between experienced and inexperienced players. *Sports Biomechanics*, *6*(2), 187–198. https://doi.org/10.1080/14763140701324966

Lees, A., Asai, T., Andersen, T. B., Nunome, H., & Sterzing, T. (2010). The biomechanics of kicking in soccer: A review. *Journal of Sports Sciences*, *28*(8), 805–817. https://doi.org/10.1080/02640414.2010.481305

Lees, A., & Nolan, L. (1998). The biomechanics of soccer: A review. *Journal of Sports Sciences*, *16*(3), 211–234. https://doi.org/10.1080/026404198366740

Levanon, J., & Dapena, J. (1998). Comparison of the kinematics of the fullinstep kick and pass kicks in soccer. *Medicine and Science in Sports and Exercise*, *30*(6), 917–927. https://doi.org/10.1249/00005768-199806000-00022

Mitschke, C., & Milani, T. L. (2014). Soccer: Detailed analysis of played passes in the UEFA Euro 2012. *International Journal of Sports Science & Coaching*, *9*(5), 1019–1031. https://doi.org/10.1260/1747-9541.9.5.1019

Nunome, H., Asai, T., Ikegami, Y., & Sakurai, S. (2002). Three-dimensional kinetic analysis of side-foot and instep soccer kicks. *Medicine & Science in Sports & Exercise*, *34*(12), 2028–2036. https://doi.org/10.1097/00005768-200212000-00025

Nunome, H., Ikegami, Y., Kozakai, R., Apriantono, T., & Sano, S. (2006a). Segmental dynamics of soccer instep kicking with the preferred and non-preferred leg. *Journal of Sports Sciences*, *24*(5), 529–541. https://doi.org/10.1080/02640410500298024

Nunome, H., Inoue, K., Watanabe, K., Iga, T., & Akima, H. (2018). Dynamics of submaximal effort soccer instep kicking. *Journal of Sports Sciences*, *36*(22), 2588–2595. https://doi.org/10.1080/02640414.2018.1470216

Nunome, H., Lake, M., Georgakis, A., & Stergioulas, L. K. (2006b). Impact phase kinematics of instep kicking in soccer. *Journal of Sports Sciences*, *24*(1), 11–22. https://doi.org/10.1080/02640410400021450

Peacock, J., & Ball, K. (2018). The influence of joint rigidity on impact efficiency and ball velocity in football kicking. *Journal of Biomechanics*, *71*, 245–250. https://doi.org/10.1016/j.jbiomech.2018.02.015

Putnam, C. A. (1991). A segment interaction analysis of proximal-to-distal sequential segment motion patterns. *Medicine & Science in Sports and Exercise*, *23*(1), 130–144. https://doi.org/10.1249/00005768-199101000-00019

Roberts, E. M., & Metcalfe, A. (1968). Mechanical analysis of kicking. In J. Wartenweiller, E. Jokl, & M. Hebbelink (Eds.), *Biomechanics I* (pp. 315–319). Karger. https://doi.org/10.1159/000387200

Rodano, R., & Tavana, R. (1993). Three-dimensional analysis of instep kick in professional soccer players. In T. Reilly, J. Clarys, & A. Stibbe (Eds.), *Science & football II* (pp. 441–448). E & FN Spon.

Scurr, J., Abbott, V., & Ball, N. (2011). Quadriceps EMG muscle activation during accurate soccer instep kicking. *Journal of Sports Sciences*, *29*(3), 247–251. https://doi.org/10.1080/02640414.2010.523085

Scurr, J., & Hall, B. (2009). The effects of approach angle on penalty kicking accuracy and kick kinematics with recreational soccer players. *Journal of Sports Sciences & Medicine*, *8*, 230–234.

Sprigings, E., Marshall, R., Elliot, B., & Jennings, L. (1994). A three-dimensional kinematic method for determining the effective of arm segment rotations in producing racquet-head speed. *Journal of Biomechanics*, *27*(3), 245–254. https://doi.org/10.1016/0021-9290(94)90001-9

Teixeira, L. (1999). Kinematics of kicking as a function of different sources of constraint on accuracy. *Perceptual and Motor Skills*, *88*(3), 785–789. https://doi.org/10.2466/pms.1999.88.3.785

Zago, M., Motta, A. F., Mapelli, A., Annoni, I., Galvani, C., & Sforza, C. (2014). Effect of leg dominance on the center-of-mass kinematics during an inside-of-the-foot kick in amateur soccer players. *Journal of Human Kinetics*, *42*(1), 51–61. https://doi.org/10.2478/hukin-2014-0060